

```
# zomato dataset project

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
df=pd.read_csv('zomato.csv',encoding='latin-1')
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Cu
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Bc
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Bc
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Bc
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Bc
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Bc

5 rows × 21 columns

```
df.isnull().sum()
```

	0
Restaurant ID	0
Restaurant Name	0
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	9
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0
Is delivering now	0
Switch to order menu	0
Price range	0
Aggregate rating	0
Rating color	0
Rating text	0
Votes	0

dtype: int64

```
df_country=pd.read_excel('Country-Code.xlsx')
df_country.head()
```

	Country Code	Country
0	1	India
1	14	Australia
2	30	Brazil
3	37	Canada
4	94	Indonesia

```
#now i want to merge the country table with the df of zomato dataset
## so syntax is pd.merge() it will have the two table names to be merged
## on ='' on whic column do u want to get merged in this case country both have
# this comman column and next att is how=''we mention which join left ,righth etc

final_df=pd.merge(df,df_country,on='Country Code',how='left')
final_df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	bo
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	
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2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	
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4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	

5 rows × 22 columns

```
country_names=final_df.Country.value_counts().index
country_names
```

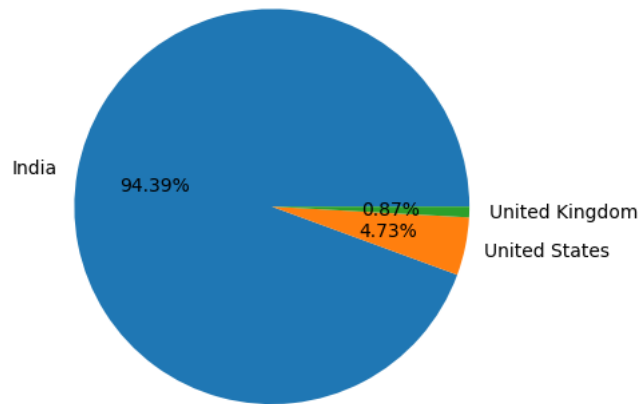
```
Index(['India', 'United States', 'United Kingdom', 'Brazil', 'South Africa',
      'UAE', 'New Zealand', 'Turkey', 'Australia', 'Phillipines', 'Indonesia',
      'Qatar', 'Singapore', 'Sri Lanka', 'Canada'],
      dtype='object', name='Country')
```

```
country_values=final_df.Country.value_counts().values
country_values
```

```
array([8652, 434, 80, 60, 60, 60, 40, 34, 24, 22, 21,
       20, 20, 20, 4])
```

```
plt.pie(country_values[:3],labels=country_names[:3],autopct='%1.2f%%')
```

```
([<matplotlib.patches.Wedge at 0x7ddefd80f6b0>,
<matplotlib.patches.Wedge at 0x7ddefd80c2c0>,
<matplotlib.patches.Wedge at 0x7ddefdbb8890>],
[Text(-1.082974277862112, 0.1927867046480056, 'India'),
Text(1.0772816964394372, -0.22240536530526556, 'United States'),
Text(1.0995865232164619, -0.030157552300104404, 'United Kingdom')],
[Text(-0.5907132424702428, 0.10515638435345759, '94.39%'),
Text(0.5876081980578747, -0.12131201743923574, '4.73%'),
Text(0.5997744672089791, -0.01644957398187513, '0.87%')])
```

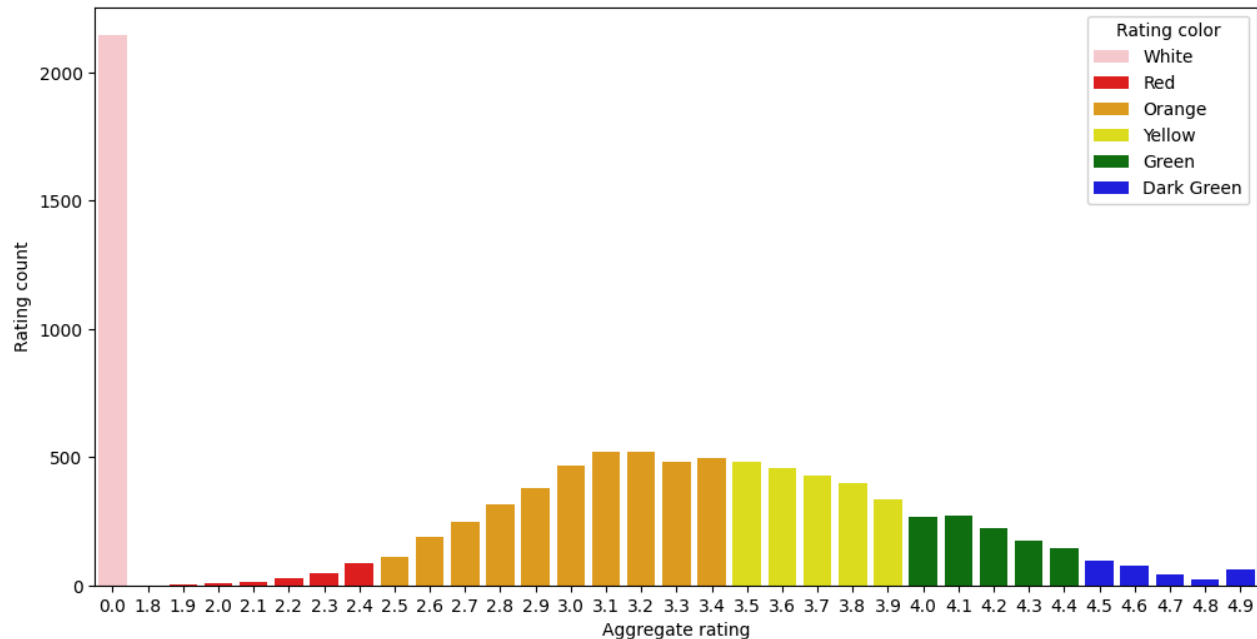


```
ratings=final_df.groupby(['Aggregate rating','Rating color','Rating text']).size().reset_index().rename(columns={0:'Rating
ratings
```

	Aggregate rating	Rating color	Rating text	Rating count
0	0.0	White	Not rated	2148
1	1.8	Red	Poor	1
2	1.9	Red	Poor	2
3	2.0	Red	Poor	7
4	2.1	Red	Poor	15
5	2.2	Red	Poor	27
6	2.3	Red	Poor	47
7	2.4	Red	Poor	87
8	2.5	Orange	Average	110
9	2.6	Orange	Average	191
10	2.7	Orange	Average	250
11	2.8	Orange	Average	315
12	2.9	Orange	Average	381
13	3.0	Orange	Average	468
14	3.1	Orange	Average	519
15	3.2	Orange	Average	522
16	3.3	Orange	Average	483
17	3.4	Orange	Average	498
18	3.5	Yellow	Good	480
19	3.6	Yellow	Good	458
20	3.7	Yellow	Good	427
21	3.8	Yellow	Good	400
22	3.9	Yellow	Good	335
23	4.0	Green	Very Good	266
24	4.1	Green	Very Good	274
25	4.2	Green	Very Good	221
26	4.3	Green	Very Good	174
27	4.4	Green	Very Good	144
28	4.5	Dark Green	Excellent	95
29	4.6	Dark Green	Excellent	78
30	4.7	Dark Green	Excellent	42
31	4.8	Dark Green	Excellent	25
32	4.9	Dark Green	Excellent	61

```
plt.figure(figsize=(12,6))
sns.barplot(x='Aggregate rating',y='Rating count',hue='Rating color',data=ratings,palette=['pink','Red','Orange','Yellow'],
```

<Axes: xlabel='Aggregate rating', ylabel='Rating count'>

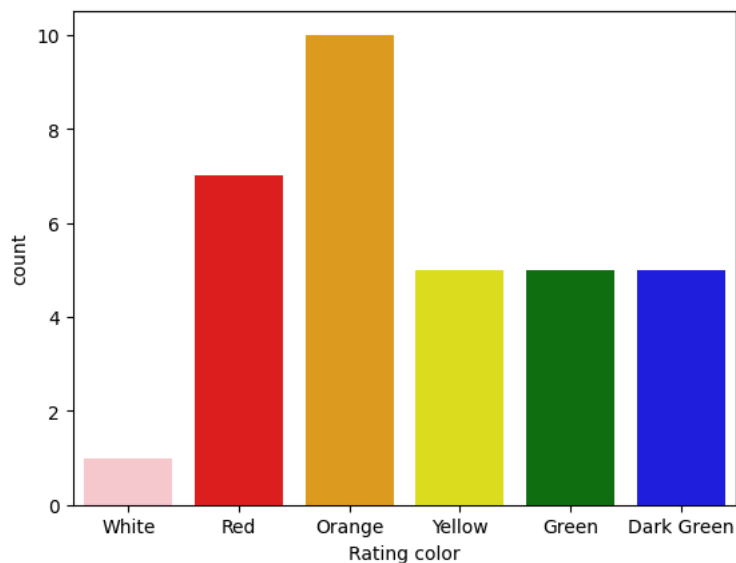


```
sns.countplot(x='Rating color',data=ratings,palette=['pink','Red','Orange','Yellow','Green','blue'])
```

/tmp/ipykernel_1507/3827354541.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and

```
sns.countplot(x='Rating color',data=ratings,palette=['pink','Red','Orange','Yellow','Green','blue'])
<Axes: xlabel='Rating color', ylabel='count'>
```



```
final_df[final_df['Rating color']=='White'].groupby('Country').size().reset_index()
```

	Country	0
0	Brazil	5
1	India	2139
2	United Kingdom	1
3	United States	3

```
final_df.groupby(['Country','Currency']).size().reset_index().rename(columns={0:'Rating count'})
```

	Country	Currency	Rating count
0	Australia	Dollar(\$)	24
1	Brazil	Brazilian Real(R\$)	60
2	Canada	Dollar(\$)	4
3	India	Indian Rupees(Rs.)	8652
4	Indonesia	Indonesian Rupiah(IDR)	21
5	New Zealand	NewZealand(\$)	40